

用于分子毒性预测的化学语言模型任务特定稀疏特征掩码

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摘要——可靠的计算机中分子毒性预测是现代药物发现的基石，提供了一种可扩展的实验筛选替代方案。然而，最先进模型的黑箱特性仍然是应用的重大障碍，因为高风险的安全决策需要可验证的结构洞察力与预测性能并重。为了解决这一问题，我们提出了一种新的多任务学习（MTL）框架，旨在同时提高准确性和可解释性。我们的架构将共享的化学语言模型与特定任务的注意力模块结合。通过对这些模块施加L1稀疏性惩罚，该框架被限制为针对每个不同的毒性端点专注于最少的一组显著分子片段。该框架可端到端训练，并可轻松适配各种基于Transformer的骨干网络。在ClinTox、SIDER和Tox21基准数据集上的评估表明，我们的方法在性能上持续优于单任务模型和

索引词——化学语言模型、计算毒理学、可解释人工智能、分子性质预测、多任务学习、稀疏性正则化

一、介绍

分子毒性评估是在早期药物发现和化学品安全筛选中至关重要但具有挑战性的一步。传统的实验测定通常耗费大量资源且耗时长，形成了显著的瓶颈。因此，开发准确且高效的计算毒理学方法已成为关键优先事项。

本研究得到香港都会大学（项目 RD/2025/1.15）和香港研究资助局（项目 UGC/FDS16/E16/23）的支持。

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毒理学用于加速潜在有害化合物的识别。

近年来，该领域越来越多地采用基于序列的深度学习方法，并在分子的SMILES表示上进行操作。化学语言模型，如ChemBERTa [1]、MoLFormer [2]和SMILES-BERT [3]，在庞大的未标记化学语料上进行预训练，在为下游预测任务学习丰富的上下文表示方面表现出了显著的成功。然而，它们的性能往往受到标注毒理学数据稀缺的制约。为了解决这一问题，多任务学习（MTL）[4]已成为广泛采用的范式 [5], [6]，通过同时从多个相关端点学习，使模型能够更好地概括。

尽管具有优势，目前主流的基于变换器的多任务学习（MTL）架构 [7], [8]通常采用硬参数共享（HPS）[9]方法，但仍面临两个关键限制。首先，HPS 缺乏用于任务特定特征选择的明确机制；来自共享骨干网络的最终潜在表示被整体传递到每个预测头，假设所有特征对每个下游任务同样重要。这通常是不理想的，因为不同的毒性机制依赖于不同的分子特性 [10]，可能导致负迁移，其中冲突的梯度会使共享表示不稳定 [11]。其次，这些模型的不透明性限制了它们在安全关键决策中的应用价值。尽管标准架构可能实现高预测准确性，但它们掩盖了决策过程，未能提供相对于底层化学结构的透明特征归因。在毒理学中，不良反应通常是由

Task-Specific Sparse Feature Masks for Molecular Toxicity Prediction with Chemical Language Models

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Abstract——Reliable *in silico* molecular toxicity prediction is a cornerstone of modern drug discovery, offering a scalable alternative to experimental screening. However, the black-box nature of state-of-the-art models remains a significant barrier to adoption, as high-stakes safety decisions demand verifiable structural insights alongside predictive performance. To address this, we propose a novel multi-task learning (MTL) framework designed to jointly enhance accuracy and interpretability. Our architecture integrates a shared chemical language model with task-specific attention modules. By imposing an L1 sparsity penalty on these modules, the framework is constrained to focus on a minimal set of salient molecular fragments for each distinct toxicity endpoint. The resulting framework is trained end-to-end and is readily adaptable to various transformer-based backbones. Evaluated on the ClinTox, SIDER, and Tox21 benchmark datasets, our approach consistently outperforms both single-task and standard MTL baselines. Crucially, the sparse attention weights provide chemically intuitive visualizations that reveal the specific fragments influencing predictions, thereby enhancing insight into the model’s decision-making process.

Index Terms——Chemical language models, computational toxicology, explainable AI, molecular property prediction, multi-task learning, sparsity regularization

I. INTRODUCTION

The assessment of molecular toxicity is a crucial yet challenging step in early-stage drug discovery and chemical safety screening. Traditional experimental assays are often resource-intensive and time-consuming, creating a significant bottleneck. Consequently, developing accurate and efficient *in silico* methods has become a key priority in computational tox-

This work is supported by Hong Kong Metropolitan University (Project RD/2025/1.15) and Hong Kong Research Grants Council (Project UGC/FDS16/E16/23).

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icology to accelerate the identification of potentially hazardous compounds.

In recent years, the field has increasingly adopted sequence-based deep learning approaches operating on SMILES representations of molecules. Chemical language models, such as ChemBERTa [1], MoLFormer [2], and SMILES-BERT [3], pre-trained on vast unlabeled chemical corpora, have shown remarkable success in learning rich, contextual representations for downstream prediction tasks. However, their performance is often constrained by the scarcity of labeled toxicological data. To address this, MTL [4] has become a widely adopted paradigm [5], [6], enabling models to generalize better by jointly learning from several related endpoints.

Despite their advantages, prevailing transformer-based MTL architectures [7], [8], which typically use a hard parameter sharing (HPS) [9] approach, face two critical limitations. First, HPS lacks an explicit mechanism for task-specific feature selection; the final latent representation from the shared backbone is passed wholesale to each prediction head, assuming all features are equally relevant to every downstream task. This is often suboptimal, as distinct toxicity mechanisms depend on different molecular properties [10], potentially leading to negative transfer where conflicting gradients destabilize the shared representation [11]. Second, the opacity of these models restricts their utility in safety-critical decision making. While standard architectures may achieve high predictive accuracy, they obscure the decision-making process, failing to provide transparent feature attribution relative to the underlying chemical structure. In toxicology, where adverse effects are frequently driven by precise molecular substructures (toxophores), this interpretability gap remains a significant barrier

整合到发现流程中。

为了弥合这一差距，我们提出了一个模型无关的多任务学习（MTL）框架，通过任务特定的特征门控同时提升性能和可解释性。我们的贡献有三方面：

1) 我们提出了一种架构，将任务特定的注意力机制（受 MTAN [12] 启发）整合到 HPS 框架中。该设计使每个毒性终端能够从共享的化学嵌入中学习定制表示，该嵌入由任何基于 Transformer 的底层骨干生成。

2) 我们在注意力掩码上引入 L1 稀疏性惩罚，以将模型的关注限制在每个任务的关键分子片段的最小子集上。这种正则化使模型的特征选择过程与毒理学直觉一致，即特定的子结构往往决定毒性反应。

3) 我们在基准毒性数据集上验证了我们的模型，展示了具有竞争力的预测性能，同时生成了化学直观的可视化。这些可视化突出了有助于专家验证的分子特征，从而提升了模型决策过程的透明度。

二、方法

我们现在介绍我们的多任务学习（MTL）框架，分析任务特定的注意力模块，并描述带有注意力分数稀疏正则化的目标函数的公式化。

A. 总体架构

图 1 提供了我们多任务学习 (MTL) 架构的概述。该模型基于 HPS 范例构建，其中一个共享的主干网络用于所有任务。在我们的实现中，该主干网络由一个预训练的 Transformer 组成，用于处理标记化的 SMILES 字符串，以生成上下文化的分子表示。

在这个共享骨干网络的基础上，分支出 K 个任务特定的头部，每个头部都包含我们提出的注意力模块。该模块学习一个任务特定的软注意力掩码，用于重新加权来自骨干网络的共享表示。因此，虽然骨干网络被训练以生成广泛适用的化学特征，但注意力模块提供了一种机制，使每个毒理学终点可以独立地重新加权这些表示。这种设计有助于共享知识与任务特定专业化的联合优化。

为了生成最终预测，将重新加权的特征汇总为每个任务的单一固定大小的上下文向量。这是通过加权池化操作实现的：对掩码特征进行求和，然后通过累积的注意力分数进行归一化以保持表示的稳定性。得到的向量随后被输入到特定任务的线性层中，以获得最终的输出 logit。

B. 任务特定注意力模块

每个预测头的核心是一个特定任务的注意力模块，如图 2 所示。其目的是动态调节共享骨干网络的输出，通过抑制无关的化学符号，为每个任务创建定制的特征表示。

让变压器骨干网络的最终隐藏层生成一个标记表示序列 $\mathbf{H} \in \mathbb{R}^{L \times D_h}$ ，其中 L 是序列长度， D_h 是隐藏维度。从结构上看，注意力模块使用一个轻量级的子网络来生成掩码分数。该子网络实现为一个两层前馈神经网络（FFNN），并包含 LayerNorm 和 GELU 激活函数。其输出经过 sigmoid 激活，以确保注意力权重在 0 到 1 之间。

任务 k 的掩码生成 $\mathbf{M}_k = \{m_{k,1}, \dots, m_{k,L}\} \in [0, 1]^L$ 可以表示为：

$$\mathbf{M}_k = \sigma(F_k(\mathbf{H})), \quad (1)$$

其中 F_k 表示特定任务的前馈神经网络 (FFNN)， σ 是 sigmoid 函数。在最后的门控步骤中，该掩码随后逐元素应用于共享特征表示 $\mathbf{H}$ ，以生成特定任务的特征集 $\hat{\mathbf{H}}_k$ ：

$$\hat{\mathbf{H}}_k = \mathbf{M}_k \odot \mathbf{H}, \quad (2)$$

其中 $\odot$ 表示按元素进行的乘法，并进行广播沿着隐藏的维度 D_h 。

C. 目标函数

我们的多任务学习目标是一个复合函数，旨在在提高预测准确性的同时，通过在注意力掩码中强制稀疏性来促进可解释性。整体目标的构建首先是为每个单独任务定义总损失，其中包括预测损失组件和正则化惩罚。然后将这些任务级别的损失取平均，形成最终的训练目标。

对于每个任务 k ，总损失 $\mathcal{L}_k$ 定义为：

$$\mathcal{L}_k = \mathcal{L}_{\text{pred},k} + \lambda \mathcal{R}_{a,k}, \quad (3)$$

其中 $\mathcal{L}_{\text{pred},k}$ 是预测损失， $\mathcal{R}_{a,k}$ 是该任务的注意力正则项， λ 是控制正则强度的超参数。

预测部分， $\mathcal{L}_{\text{pred},k}$ ，是二元交叉熵（BCE）损失。对于包含 N 个分子的数据库，它的计算方式如下：

$$\mathcal{L}_{\text{pred},k} = \frac{1}{\sum_{n=1}^N \delta_{n,k}} \sum_{n=1}^N \delta_{n,k} \cdot \mathcal{L}_{\text{BCE}}(\sigma(\hat{y}_{n,k}), y_{n,k}), \quad (4)$$

其中 $y_{n,k}$ 和 $\hat{y}_{n,k}$ 分别是第 n 个样本和第 k 个任务的真实标签和预测 logits，并且 σ 表示 sigmoid 函数。指标 $\delta_{n,k}$ 为 1 表示样本 n 在任务 k 上存在有效标签，否则为 0，从而确保损失只在有标签的样本上计算。

正则化组件 $\mathcal{R}_{a,k}$ 是应用于注意力掩码的 L1 惩罚，以鼓励稀疏性。这会迫使

to integration into discovery pipelines.

To bridge this gap, we introduce a model-agnostic MTL framework that simultaneously enhances performance and interpretability via task-specific feature gating. Our contributions are three-fold:

- 1) We propose an architecture that adapts task-specific attention mechanisms (inspired by MTAN [12]) into an HPS framework. This design enables each toxicity endpoint to learn a tailored representation from a shared chemical embedding generated by any underlying transformer-based backbone.
- 2) We introduce an L1 sparsity penalty on the attention masks to constrain the model’s focus to a minimal subset of salient molecular fragments for each task. This regularization aligns the model’s feature selection process with the toxicological intuition that specific substructures often drive toxic responses.
- 3) We validate our model on benchmark toxicity datasets, demonstrating competitive predictive performance while generating chemically intuitive visualizations. These visualizations highlight molecular features that facilitate expert verification, thus enhancing the transparency of the model’s decision-making.

II. METHODS

We now introduce our MTL framework, dissect the task-specific attention modules, and describe the formulation of the objective function with sparsity regularization on the attention scores.

A. Overall Architecture

Fig. 1 provides an overview of our MTL architecture. The model is built upon a HPS paradigm, where a single, shared backbone is utilized across all tasks. In our implementation, this backbone consists of a pre-trained transformer that processes tokenized SMILES strings to generate contextualized molecular representations.

Branching from this shared backbone are K task-specific heads, each incorporating our proposed attention module. This module learns a task-specific soft attention mask that re-weights the shared representations from the backbone. Thus, while the backbone is trained to generate broadly applicable chemical features, the attention modules provide a mechanism to independently re-weight these representations for each toxicological endpoint. This design facilitates a joint optimization of shared knowledge and task-specific specialization.

To generate the final prediction, the re-weighted features are aggregated into a single, fixed-size context vector for each task. This is achieved via a weighted pooling operation: the masked features are summed and subsequently normalized by the accumulated attention scores to maintain representation stability. The resulting vector is then fed into a task-specific linear layer to obtain the final output logit.

B. Task-Specific Attention Module

The core of each prediction head is a task-specific attention module, as depicted in Fig. 2. Its purpose is to dynamically modulate the output of the shared backbone, creating a tailored feature representation for each task by suppressing irrelevant chemical tokens.

Let the final hidden layer of the transformer backbone produce a sequence of token representations $\mathbf{H} \in \mathbb{R}^{L \times D_h}$, where L is the sequence length and D_h is the hidden dimension. Structurally, the attention module utilizes a lightweight sub-network to generate the mask scores. This sub-network is implemented as a two-layer feed-forward neural network (FFNN) with a LayerNorm and a GELU activation. Its output is passed through a sigmoid activation to ensure the attention weights are bounded between 0 and 1.

The generation of the mask for task k , $\mathbf{M}_k = \{m_{k,1}, \dots, m_{k,L}\} \in [0, 1]^L$, can be expressed as:

$$\mathbf{M}_k = \sigma(F_k(\mathbf{H})), \quad (1)$$

where F_k represents the task-specific FFNN and σ is the sigmoid function. In the final gating step, this mask is then applied element-wise to the shared feature representation $\mathbf{H}$ to produce a task-specific feature set $\hat{\mathbf{H}}_k$:

$$\hat{\mathbf{H}}_k = \mathbf{M}_k \odot \mathbf{H}, \quad (2)$$

where $\odot$ denotes element-wise multiplication that is broadcast along the hidden dimension D_h .

C. Objective Function

Our MTL objective is a composite function designed to optimize predictive accuracy while promoting interpretability by enforcing sparsity in the attention masks. The overall objective is constructed by first defining a total loss for each individual task, which includes both a prediction loss component and a regularization penalty. These task-level losses are then averaged to form the final training objective.

For each task k , the total loss $\mathcal{L}_k$ is given by:

$$\mathcal{L}_k = \mathcal{L}_{\text{pred},k} + \lambda \mathcal{R}_{a,k}, \quad (3)$$

where $\mathcal{L}_{\text{pred},k}$ is the prediction loss, $\mathcal{R}_{a,k}$ is the attention regularization term for that task, and λ is a hyperparameter controlling the regularization strength.

The prediction component, $\mathcal{L}_{\text{pred},k}$, is the Binary Cross-Entropy (BCE) loss. For a dataset of N molecules, it is computed as:

$$\mathcal{L}_{\text{pred},k} = \frac{1}{\sum_{n=1}^N \delta_{n,k}} \sum_{n=1}^N \delta_{n,k} \cdot \mathcal{L}_{\text{BCE}}(\sigma(\hat{y}_{n,k}), y_{n,k}), \quad (4)$$

where $y_{n,k}$ and $\hat{y}_{n,k}$ are the ground truth label and predicted logit for the n -th sample and k -th task, respectively, and σ denotes the sigmoid function. The indicator $\delta_{n,k}$ is 1 if a valid label exists for sample n on task k and 0 otherwise, ensuring that the loss is computed only over labeled samples.

The regularization component, $\mathcal{R}_{a,k}$, is an L1 penalty applied to the attention masks to encourage sparsity. This forces

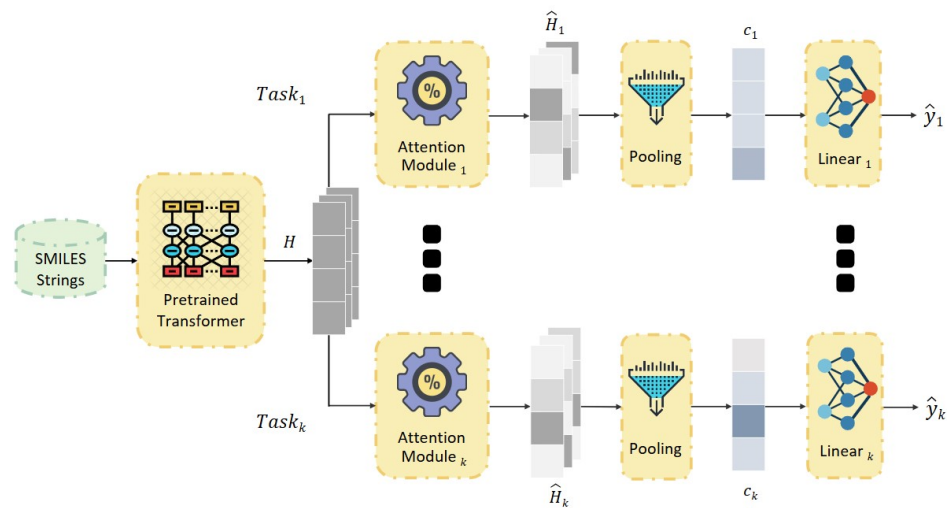


图 1: 整体多任务学习架构。一个共享的 Transformer 主干将 SMILES 字符串处理成无上下文的化学表示 (H)。该表示被分配到 K 个任务特定的头。在每个头中, 一个注意力模块 (图 2) 生成任务特定的特征集 (H_k), 然后通过加权池化聚合为固定大小的上下文向量 (c_k) 用于最终预测 (y_k)。

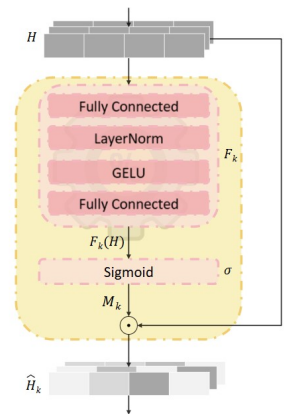


图2: 任务特定注意力模块。共享特征 (H) 通过前馈网络 (F_k) 和 Sigmoid 激活函数 (σ) 处理, 以生成一个软掩码 (M_k)。该掩码通过逐元素相乘 ($\odot$) 应用于原始输入, 生成重加权的、任务特定的特征表示 (H_k), 其中较暗区域表示更高的显著性。

该模型专注于一组最具显著性的关键标记, 从而隔离出对预测影响最大的特定分子片段。该惩罚在任务 k 的所有有效样本中取平均值:

$$\mathcal{R}_{a,k} = \frac{1}{\sum_{n=1}^N \delta_{n,k}} \sum_{n=1}^N \delta_{n,k} \|\mathbf{M}_{n,k}\|_1, \quad (5)$$

其中 $\mathbf{M}_{n,k}$ 是由第 k 个任务的注意力模块为第 n 个样本生成的注意力掩码 (1)。

最终的总体目标函数 $\mathcal{L}$ 是这些的平均值

任务级损失:

$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_k = \frac{1}{K} \sum_{k=1}^K (\mathcal{L}_{\text{pred},k} + \lambda \mathcal{R}_{a,k}). \quad (6)$$

三、实验

我们在分子毒性基准上评估我们的框架, 以验证预测性能和可解释性。我们的实验涵盖了不同规模和任务复杂度的数据集。具体来说, 我们在三个 MoleculeNet 基准上进行实验: ClinTox (1,484种化合物, 2个任务)、SIDER (1,427种药物, 27个不良反应任务) 以及 Tox21 (7,831种化合物, 12个受体检测任务)。

我们通过 HuggingFace [17] 实例化了三个预训练的 Transformer 主干模型。为了评估在专业化和通用架构下的性能, 我们选择了 ChemBERTa-1 [1], 和 MoLFormer [2] 作为代表性的化学语言模型, 同时将 DistilBERT [18] 作为通用 NLP 基线模型。我们提出的稀疏注意力多任务学习框架 (MTL-SA) 与单任务学习 (STL) 以及采用硬参数共享的标准多任务学习 (MTL-HPS) [7], [8] 进行比较。为了确保公平比较, 所有模型均使用相同的数据预处理和超参数优化流程进行训练。我们首先介绍实验设置, 然后展示基准结果、稀疏正则化的消融研究, 以及对学习到的注意力模式的定性分析。

A. 实验装置

使用 RDKit 对原始 SMILES 字符串进行验证和规范化。每个数据集通过迭代分层抽样 [19] 划分为 80% 的训练集和 20% 的测试集。超参数优化采用带有贝叶斯算法的 5 折交叉验证

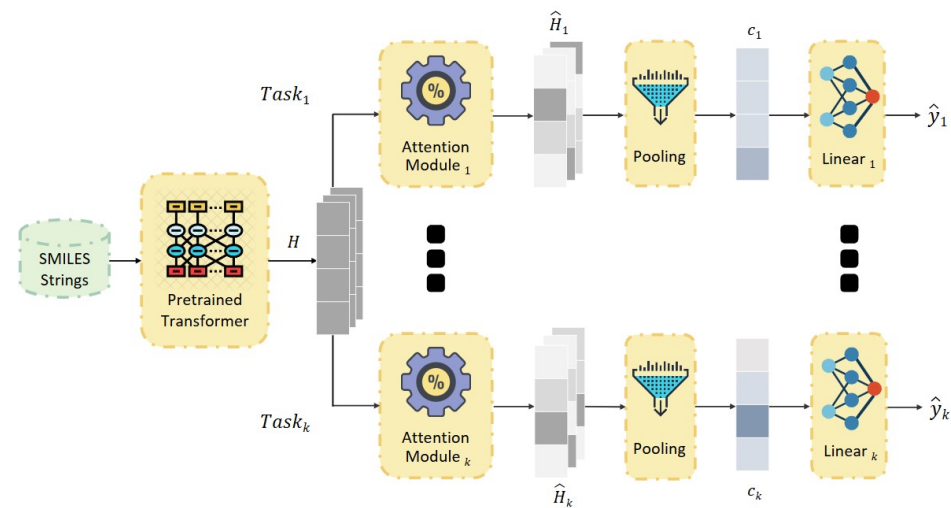


Fig. 1: **Overall Multi-Task Learning Architecture.** A shared transformer backbone processes SMILES strings into a contextual chemical representation (H). This representation is distributed to K task-specific heads. In each head, an attention module (Fig. 2) generates a task-specific feature set (H_k), which is then aggregated via weighted pooling into a fixed-size context vector (c_k) for final prediction (y_k).

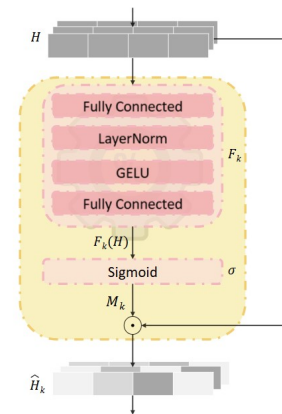


Fig. 2: **Task-Specific Attention Module.** Shared features (H) are processed by a feed-forward network (F_k) and a Sigmoid activation (σ) to generate a soft mask (M_k). This mask is applied to the original input via element-wise multiplication ($\odot$), producing a re-weighted, task-specific feature representation (H_k) where darker regions indicate higher salience.

the model to focus on a minimal set of highly salient tokens, hence isolating the specific molecular fragments that most significantly influence the prediction. This penalty is averaged across all valid samples for task k :

$$\mathcal{R}_{a,k} = \frac{1}{\sum_{n=1}^N \delta_{n,k}} \sum_{n=1}^N \delta_{n,k} \|\mathbf{M}_{n,k}\|_1, \quad (5)$$

where $\mathbf{M}_{n,k}$ is the attention mask (1) generated for the n -th sample by the k -th task's attention module.

The final, overall objective function $\mathcal{L}$ is the mean of these

task-level losses:

$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_k = \frac{1}{K} \sum_{k=1}^K (\mathcal{L}_{\text{pred},k} + \lambda \mathcal{R}_{a,k}). \quad (6)$$

III. EXPERIMENTS

We evaluate our framework on molecular toxicity benchmarks to validate both predictive performance and interpretability. Our experiments span datasets of varying scale and task complexity. Specifically, we conduct experiments on three MoleculeNet [13] benchmarks: ClinTox (1,484 compounds, 2 tasks) [14], SIDER (1,427 drugs, 27 adverse reaction tasks) [15], and Tox21 (7,831 compounds, 12 receptor assays) [16].

We instantiated three pre-trained transformer backbones via HuggingFace [17]. To assess performance across both specialized and general architectures, we selected ChemBERTa-1 [1], and MoLFormer [2] as representative chemical language models, while including DistilBERT [18] as a general-purpose NLP baseline. Our proposed MTL framework with sparse attention (MTL-SA) is compared against single-task learning (STL) and standard multi-task learning with hard parameter sharing (MTL-HPS) following [7], [8]. All models are trained using identical preprocessing and hyperparameter optimization procedures to ensure fair comparison. We first describe our experimental setup before presenting benchmark results, ablation studies on sparsity regularization, and qualitative analysis of learned attention patterns.

A. Experimental Setup

Raw SMILES strings are validated and canonicalized using RDKit. Each dataset is partitioned into 80% training and 20% test sets via iterative stratified sampling [19]. Hyperparameter optimization employs 5-fold cross-validation with Bayesian

使用 Optuna [20] 进行搜索。最终模型在完整的 80% 训练集上重新训练，并在保留的 20% 测试集上进行一次评估。性能使用所有任务的宏平均 ROC-AUC 来衡量。

所有模型都使用 PyTorch 2.6.0 在 NVIDIA A800 GPU 上通过 AdamW [21] 微调。

B. 结果

本节展示了不同学习范式下的基准性能比较，随后进行消融研究，考察 L1 稀疏正则化强度的影响。

1) 基准性能：表 I 报告了所有学习框架下的 ROC-AUC 分数。经典的机器学习模型建立了强劲的初始基线，其中随机森林在 SIDER 数据集上尤为突出，取得了 0.6581 的竞争性分数。如预期，基于 Transformer 的模型总体上表现更优，尤其是在 ClinTox 和 Tox21 数据集上。然而，标准的多任务超参数搜索 (MTL-HPS) 表现参差不齐；在某些情况下，它比单任务学习 (STL) 有所提升，但在其他情况下会导致性能下降（例如，ChemBERTa-1 在 ClinTox 和 SIDER 上），这一现象很可能归因于负迁移。

相比之下，我们提出的方法在几乎所有主干网络-数据集组合中，都始终优于 STL 和 MTL-HPS 基线。当使用专门的化学语言模型时，尤其是 MoLFormer 主干网络，我们的方法优势最为明显，在 ClinTox (0.9482) 和 Tox21 (0.8528) 上取得了最高的整体分数，并在所有测试模型中获得了最佳平均性能 (0.8149)。这种持续的稳定性表明，我们的任务专用门控在一定程度上隐式缓解了负迁移问题。通过使每个头能够过滤掉共享主干网络中不相关或冲突的特征，模型减少了通常会影响标准 HPS 架构的干扰。

表 I: 基准数据集上的性能比较 (ROC-AUC)。我们提出的 MTL-SA 与经典机器学习基线、STL 和 MTL-HPS 在三种 Transformer 主干模型上进行比较。每个主干模型-数据集组中的最高分以粗体标出。每个数据集在所有方法中取得的最高总体分数用下划线标出。

模型		性能			
		(2 个任务)	(27 任务)	(12 任务)	平均
机器学习	临床毒理	侧面	Tox21		
	ECFP+RF	0.8629	0.6581	0.8199	0.7803
	ECFP+支持向量机	0.8657	0.6462	0.8057	0.7725
	ECFP+KNN	0.7079	0.5992	0.7300	0.6790
学习		性能			
支柱	框架	临床毒理	侧面	Tox21	平均
	STL	0.9329	0.6036	0.8157	0.7841
变形金刚	MTL-HPS	0.9360	0.6076	0.8419	0.7952
	MTL-SA (我们的)	0.9370	0.6231	0.8439	0.8013
	STL	0.9202	0.6122	0.8188	0.7837
ChemBERTa-1	MTL-HPS	0.9194	0.6015	0.8378	0.7862
	MTL-SA (我们的)	0.9234	0.6202	0.8362	0.7933
	STL	0.9429	0.6239	0.8367	0.8012
MoLFormer	MTL-HPS	0.9390	0.6321	0.8475	0.8062
	MTL-SA (我们的)	0.9482	0.6438	0.8528	0.8149

2) 正则化消融实验：表 II 使用固定超参数在 MoLFormer 骨干网络上评估 L1 正则化强度 λ ，以隔离稀疏性正则化的影响。与担心施加稀疏性可能过度限制特征选择并降低准确性的顾虑相反，在我们的框架中应用适当的正则化强度 ($\lambda = 10^{-3}$) 带来双重好处：它既有助于模型可解释性，同时相对于未正则化的基线 ($\lambda = 0$) 提升了预测性能。在 ClinTox、SIDER 和 Tox21 数据集上，中等强度的正则化分别带来了 +0.36%、+0.23% 和 +0.38% 的提升。

表 II: 关于 L1 稀疏正则强度 (λ) 的消融研究。报告了 MoLFormer 在不同超参数值 (λ) 下的 ROC-AUC 性能。百分比变化 (Δ) 是相对于未正则化基线 ($\lambda = 0$) 计算的。

λ	临床毒理		侧面		Tox21	
	ROC-AUC	Δ (%)	ROC-AUC	Δ (%)	ROC-AUC	Δ (%)
0	0.9439	–	0.6184	–	0.8395	–
10^{-4}	0.9427	-0.12%	0.6194	+0.16%	0.8393	-0.02%
10^{-3}	0.9473	+0.36%	0.6198	+0.23%	0.8427	+0.38%
10^{-2}	0.9464	+0.26%	0.6169	-0.24%	0.8412	+0.20%

C. 通过注意力模式进行可解释性

MoLFormer 的分词策略本质上支持原子级分析 [2]。我们利用这一点来检查我们的稀疏性约束是否成功地隔离了化学上相关的片段。通过可视化学习到的注意力掩码，我们观察到模型识别的是可化学解释的特征，而不是依赖于扩散或无关的背景信息。我们使用 Tox21 和 SIDER 数据集中的示例来说明这一能力，这有效地展示了两个关键行为：任务特定的区分能力和跨骨架的一致性。

首先，该模型展示了根据目标毒性终点关注同一分子中不同特征的能力。如图3所示，对于NR-PPAR- γ 任务（左图），注意力机制对乙烯基片段上的溴 (Br) 原子给予了较高权重。相反，对于SR-ATAD5任务（右图），注意力则完全转向羧酸盐基团的阴离子氧 (O^-)。这种区分表明，任务特定的门控机制能够有效地过滤共享表示，使每个注意力头能够专注于被认为与其生物学终点最相关的特定原子特征。

其次，该模型在结构多样的分子中表现出特征选择的一致性，这一特性对于稳健的泛化能力至关重要。在 SIDER 血管疾病任务（图 4）中，尽管存在显著的结构差异，该模型仍能持续突出氮原子。具体而言，它在地西洋（左图）的苯二氮卓环中识别出酰胺氮，在美哌啶（右图）的苯基哌啶骨架中识别出哌啶氮。这表明

search using Optuna [20]. Final models are retrained on the full 80% training set and evaluated once on the held-out 20% test set. Performance is measured using macro-averaged ROC-AUC across all tasks.

All models are fine-tuned with AdamW [21] on NVIDIA A800 GPUs using PyTorch 2.6.0.

B. Results

This section presents benchmark performance comparisons across learning paradigms, followed by ablation studies examining the effect of L1 sparsity regularization strength.

1) *Benchmark Performance*: Table I reports the ROC-AUC scores across all learning frameworks. Classical machine learning models establish strong initial baselines, with Random Forest notably achieving a competitive score of 0.6581 on SIDER. As expected, transformer-based models generally deliver superior performance, particularly on the ClinTox and Tox21 datasets. However, standard MTL-HPS exhibits mixed efficacy; while it improves over STL in some cases, it leads to performance degradation in others (e.g., ChemBERTa-1 on ClinTox and SIDER), a phenomenon likely attributable to negative transfer.

In contrast, our proposed method consistently outperforms both STL and MTL-HPS baselines across nearly all backbone-dataset combinations. The advantage is most pronounced when using specialized chemical language models, particularly the MoLFormer backbone, where our approach achieves the highest overall scores on ClinTox (0.9482) and Tox21 (0.8528), and secures the best average performance (0.8149) across all tested models. This consistent stability indicates that our task-specific gating implicitly mitigates negative transfer. By enabling each head to filter out irrelevant or conflicting features from the shared backbone, the model reduces the interference that typically hampers standard HPS architectures.

TABLE I: **Performance Comparison (ROC-AUC) on Benchmark Datasets.** Our proposed MTL-SA is compared against classical machine learning baselines, STL, and MTL-HPS across three transformer backbones. The highest score within each backbone-dataset group is marked in bold. The highest overall score achieved for each dataset across all methods is underlined.

		Performance				
Machine Learning	Model	(2 Tasks)	(27 Tasks)	(12 Tasks)		
		ClinTox	SIDER	Tox21	Average	
	ECFP+RF	0.8629	0.6581	0.8199	0.7803	
	ECFP+SVM	0.8657	0.6462	0.8057	0.7725	
	ECFP+KNN	0.7079	0.5992	0.7300	0.6790	
Transformers	Backbone	Performance				
	Learning Framework	ClinTox	SIDER	Tox21	Average	
	DistilBERT	STL	0.9329	0.6036	0.8157	0.7841
		MTL-HPS	0.9360	0.6076	0.8419	0.7952
		MTL-SA (Ours)	0.9370	0.6231	0.8439	0.8013
	ChemBERTa-1	STL	0.9202	0.6122	0.8188	0.7837
		MTL-HPS	0.9194	0.6015	0.8378	0.7862
		MTL-SA (Ours)	0.9234	0.6202	0.8362	0.7933
	MoLFormer	STL	0.9429	0.6239	0.8367	0.8012
		MTL-HPS	0.9390	0.6321	0.8475	0.8062
		MTL-SA (Ours)	0.9482	0.6438	0.8528	0.8149

2) *Regularization Ablation*: Table II evaluates L1 regularization strength λ using the MoLFormer backbone with fixed hyperparameters to isolate the effect of sparsity regularization. Contrary to the concern that imposing sparsity might overly constrain feature selection and degrade accuracy, applying an appropriate regularization strength ($\lambda = 10^{-3}$) in our framework yields dual benefits: it facilitates model interpretability, while simultaneously improving predictive performance relative to the unregularized baseline ($\lambda = 0$). On ClinTox, SIDER and Tox21, moderate regularization achieves gains of +0.36%, +0.23% and +0.38% respectively.

TABLE II: **Ablation Study on L1 Sparsity Regularization Strength (λ).** ROC-AUC performance with MoLFormer is reported for varying values of the hyperparameter λ . The percentage change (Δ) is calculated relative to the unregularized baseline ($\lambda = 0$).

λ	ClinTox		SIDER		Tox21	
	ROC-AUC	Δ (%)	ROC-AUC	Δ (%)	ROC-AUC	Δ (%)
0	0.9439	–	0.6184	–	0.8395	–
10^{-4}	0.9427	-0.12%	0.6194	+0.16%	0.8393	-0.02%
10^{-3}	0.9473	+0.36%	0.6198	+0.23%	0.8427	+0.38%
10^{-2}	0.9464	+0.26%	0.6169	-0.24%	0.8412	+0.20%

C. Interpretability via Attention Patterns

The tokenization strategy of MoLFormer inherently supports atom-level analysis [2]. We leverage this to examine whether our sparsity constraint successfully isolates chemically relevant fragments. By visualizing the learned attention masks, we observe that the model identifies chemically interpretable features rather than relying on diffuse or irrelevant background context. We illustrate this capability using examples from the Tox21 and SIDER datasets, which effectively demonstrate two key behaviors: task-specific differentiation and cross-scaffold consistency.

First, the model demonstrates the ability to attend to distinct features within the same molecule depending on the target toxicity endpoint. As shown in Fig. 3, for the NR-PPAR- γ task (left), the attention mechanism heavily weights the bromine (Br) atom on the vinyl fragment. In contrast, for the SR-ATAD5 task (right), attention shifts exclusively to the anionic oxygen (O^-) of the carboxylate group. This differentiation suggests that the task-specific gating mechanism effectively filters the shared representation, allowing each head to focus on specific atomic features deemed most relevant to its biological endpoint.

Second, the model exhibits consistency in feature selection across structurally diverse molecules, a property essential for robust generalization. In the SIDER Vascular Disorders task (Fig. 4), the model consistently highlights nitrogen atoms despite significant structural differences. Specifically, it isolates the amide nitrogen within the benzodiazepine ring of Diazepam (left) and the piperidine nitrogen within the phenylpiperidine scaffold of Meperidine (right). This indicates

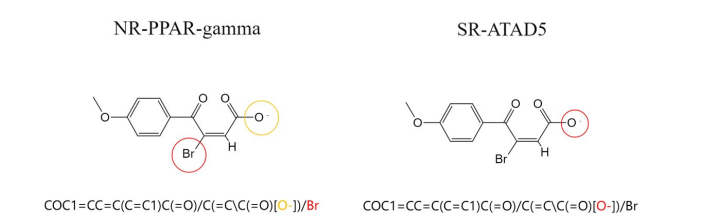


图 3: Tox21 上的任务特定区分。模型在同一分子中为不同的端点隔离出不同的特征。对于 NR-PPAR- γ 任务（左），注意力集中在溴（Br）原子上，而对于 SR-ATAD5 任务（右），注意力则集中在阴性氧（O⁻）上。具有高注意力权重的显著片段用红色标出。

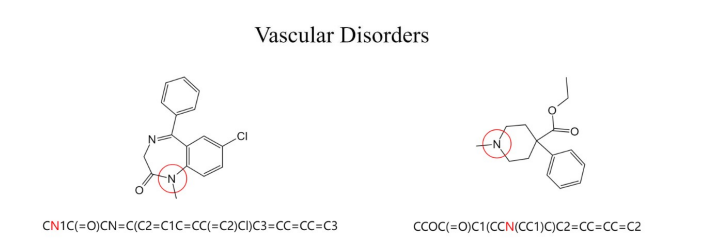


图 4: SIDER 上骨架间的毒性基团泛化。在血管疾病任务中，该模型能够在结构多样的药物中持续识别含氮基团。它突出了地西洋（左图）中的酰胺氮和哌替啶（右图）中的哌啶氮，提示模型学习的是抽象的化学原理，而不是死记硬背。

稀疏性惩罚约束模型优先考虑活性化合物中共有的基于氮的基元，而不是记忆特定的分子骨架。

这些可视化表明，L1 稀疏性惩罚不仅仅是减少噪声，而是积极促使模型学习可泛化的结构模式。通过隔离一小部分权重极高的词元，该框架为其预测提供了透明的基础，使领域专家能够评估决策过程的化学合理性。

四、结论

在这项工作中，我们提出了一个多任务学习（MTL）框架，通过整合任务特定的稀疏注意力来增强分子毒性预测。在多个 MoleculeNet 基准测试中验证，我们的方法始终优于单任务和标准多任务学习的基线方法。关键是，我们的研究表明，施加适度的 L1 稀疏正则化可以同时提高预测准确性和模型可解释性，这挑战了传统上认为两者之间存在权衡的观点。学习到的注意力模式突出显示了重要的分子片段，表现出任务特异性的区分能力和跨分子的一般化能力。通过隔离这些离散的结构 motif，该框架提供了一种

为其预测提供可解释的理由，从而能够验证模型的关注点是否与化学直觉一致。未来的工作可以探索明确的任务加权策略，以进一步减轻负迁移，并将该框架扩展 to 处理大规模异构数据问题。通过将高性能预测与透明的特征选择相结合，我们的方法代表了朝着更可靠、更可解释的计算毒理学迈出的有前景的一步。

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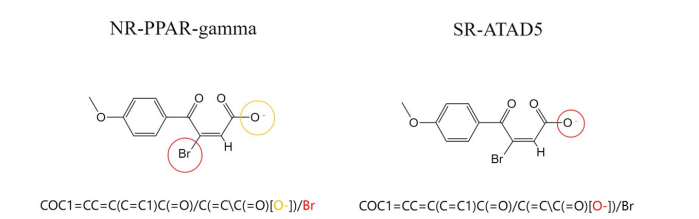


Fig. 3: **Task-Specific Differentiation on Tox21.** The model isolates distinct features within the same molecule for different endpoints. For the NR-PPAR-gamma task (left), attention targets the bromine (Br) atom, whereas for the SR-ATAD5 task (right), it focuses on the anionic oxygen (O⁻). Salient fragments with high attention weights are highlighted in red.

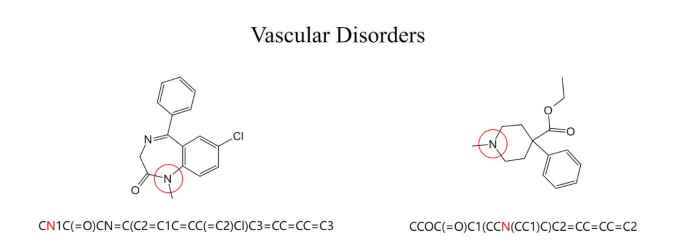


Fig. 4: **Toxicophore Generalization across Scaffolds on SIDER.** For the Vascular Disorders task, the model consistently identifies nitrogen-based motifs across structurally diverse drugs. It highlights the amide nitrogen in Diazepam (left) and the piperidine nitrogen in Meperidine (right), suggesting the learning of abstract chemical principles rather than rote memorization.

that the sparsity penalty constrains the model to prioritize common nitrogen-based motifs shared among active compounds, rather than memorizing specific molecular scaffolds.

These visualizations indicate that the L1 sparsity penalty does not merely reduce noise, but actively encourages the model to learn generalizable structural patterns. By isolating a minimal set of highly weighted tokens, the framework provides a transparent basis for its predictions, allowing domain experts to assess the chemical plausibility of the decision-making process.

IV. CONCLUSIONS

In this work, we introduced a MTL framework that enhances molecular toxicity prediction by integrating task-specific sparse attention. Validated across multiple MoleculeNet benchmarks, our approach consistently outperforms both single-task and standard MTL baselines. Crucially, our findings demonstrate that imposing moderate L1 sparsity regularization can simultaneously improve predictive accuracy and model interpretability, challenging the conventional view of a trade-off between these objectives. The learned attention patterns highlight salient molecular fragments, exhibiting task-specific discrimination and cross-molecule generalization. By isolating these discrete structural motifs, the framework offers an

interpretable rationale for its predictions, enabling validation of whether the model’s focus aligns with chemical intuition. Future work could explore explicit task-weighting strategies to further mitigate negative transfer and extend the framework to tackle large-scale, heterogeneous data problems [22]. By bridging high-performance prediction with transparent feature selection, our approach represents a promising step towards more reliable and interpretable *in silico* toxicology.

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